INCH-POUND
MIL-DTL-32707
9 May 2022

DETAIL SPECIFICATION

MITTEN SHELLS, FULL AND TRIGGER FINGER, EXTREME COLD WEATHER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers full and trigger finger extreme cold weather mitten shells.

1.2 Classification. The mitten shells are available in the following types and sizes, as specified in the contract or order (see 6.2). Requirements and verification criteria cited herein are applicable to both types.

1.2.1 Types.

Type I – Full mitten shell

Type II – Trigger finger mitten shell

1.2.2 Sizes.

Small

Medium

Large

Extra Large

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-52080 - Tape, Lacing and Tying, Nylon

A-A-59826 - Thread, Nylon

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5038 - Tape, Textile and Webbing, Textile, Reinforcing, Nylon

MIL-W-5664 - Webbing, Textile, Elastic

Comments, suggestions, or questions on this document should be addressed to Marine Corps Systems Command, 2200 Lester Street, Quantico, VA 22134 ATTN: SEAL-SE-STDS or emailed to USMC_STDZ@usmc.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 8440

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MIL-PRF-17337	-	Webbing, Textile, Woven, Nylon
MIL-DTL-32075	-	Label: for Clothing, Equipage, and Tentage, (General Use)
MIL-PRF-32567	-	Batting, Synthetic Fibers, Unquilted
MIL-DTL-32698	-	Laminate, Nylon Taslan
MIL-DTL-32699	-	Cloth, 70 Denier Nylon Ripstop
MIL-DTL-32708	-	Liner, Trigger Finger Mitten, Extreme Cold Weather

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP1	-	Gray Scale for Color Change
AATCC EP9	-	Visual Assessment of Color Difference of Textiles
AATCC TM118	-	Oil Repellency: Hydrocarbon Resistance Test

(Copies of these documents are available online at www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

ASTM INTERNATIONAL

ASTM D1610	-	Standard Practice for Conditioning Leather and Leather Products for Testing
ASTM D1776/D1776M	-	Standard Practice for Conditioning and Testing Textiles
ASTM D1814	-	Standard Test Method for Measuring Thickness of Leather Units
ASTM D1912	-	Standard Test Method for Cold-Crack Resistance of Upholstery Leather
ASTM D2209	-	Standard Test Method for Tensile Strength of Leather
ASTM D2617	-	Standard Test Method for Total Ash in Leather
ASTM D2807	-	Standard Test Method for Chromic Oxide in Leather (Perchloric Acid Oxidation)
ASTM D2810	-	Standard Test Method for pH of Leather
ASTM D2821	-	Standard Test Method for Measuring the Relative Stiffness of Leather by Means of a Torsional Wire Apparatus
ASTM D3495	-	Standard Test Method for Hexane Extraction of Leather
ASTM D4705	-	Standard Test Method for Stitch Tear Strength of Leather, Double Hole
ASTM D5052	-	Standard Test Method for Permeability of Leather to Water Vapor
ASTM D6076	-	Standard Test Method for Shrinkage Temperature of Leather

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ASTM D6182 - Standard Test Method for Flexibility and Adhesion of Finish on Leather

ASTM D6193 - Standard Practice for Stitches and Seams

ASTM D7255 - Standard Test Method for Abrasion Resistance of Leather (Rotary Platform, Abraser Method)

ASTM F3426 - Standard Test Method for Measuring the Thermal Insulation of Clothing Items Using Heated Manikin Body Forms

(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION

ISO 11640 - Leather — Tests for Colour Fastness — Colour Fastness to Cycles of To-and-Fro Rubbing

ISO 11644 - Leather — Test for Adhesion of Finish

ISO 17075-2 - Leather — Chemical Determination of Chromium(VI) Content in Leather — Part 2: Chromatographic Method

(Copies of these documents are available online at www.iso.org.)

NATIONAL FIRE PROTECTION ASSOCIATION

NFPA 1971 - Standard on Protective Ensembles for Structural Fire Fighting and Proximity Fire Fighting

(Copies of this document are available online at www.nfpa.org.)

SAE INTERNATIONAL

AMS-STD-595 - Colors Used in Government Procurement

(Copies of this document are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance inspection. When specified (see 6.2), a sample shall be subjected to a conformance inspection in accordance with 4.3.

3.2 Standard sample. The finished type I and II mitten shells and designs shall match the standard samples (see 6.4) for design and all other characteristics or acceptance criteria for which a standard sample is referenced.

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

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3.4 Design.

3.4.1 Type I, full mitten shell. The full mitten shell is an outer mitten designed to be worn in combination with a liner in extreme cold weather environments. The shell is part of a system that includes a removable insulated trigger finger mitten liner conforming to MIL-DTL-32708 that can be worn over a glove insert if desired. The outside shell is constructed from a nylon taslan backed by a waterproof layer, with a water-repellent treatment on the face conforming to 3.5.1 and a water-repellent goatskin leather palm conforming to 3.5.2, which is backed by a waterproof layer. The suede side of the leather is used to construct a nose wipe on the back of the thumb. The outer shell fabric's inside seams are seam taped to provide waterproofing. The mitten shell is insulated on the back of the hand and thumb as specified in 3.5.4; the palm, back of the hand, gauntlet, and finger/thumb portions are lined with 70 denier (D) nylon ripstop conforming to 3.5.3. This inside lining is tacked to the shell at several locations to ensure the lining material does not separate from the shell. The mitten shell has an elasticized inner wrist and an adjustable strap at the back of the wrist to provide a close fit, with a long gauntlet that extends up the forearm with a manual adjustment feature at the end of the cuff to provide desired closure over various clothing layers. An adjustable lanyard is attached to the hem, and the outside of the shell has a feature at the side seam that enables the mittens to be clipped together as a pair when not in use (see 3.5.13).

3.4.2 Type II, trigger finger mitten shell. The trigger finger mitten shell is an outer mitten designed to be worn in combination with a liner in extreme cold weather environments when enhanced dexterity is needed as compared to a full mitten shell. The trigger finger design enables the wearer to move their first finger independently from the rest of the hand and, when needed, move their first finger into the larger finger compartment to warm up. The shell is part of a system that includes a removable insulated trigger finger mitten liner conforming to MIL-DTL-32708 that can be worn over a glove insert if desired. The outside shell is constructed from a nylon taslan backed by a waterproof layer, with a water repellent treatment on the face conforming to 3.5.1 and a water repellent goatskin leather palm conforming to 3.5.2, which is backed by a waterproof layer. The suede side of the leather is used to construct a nose wipe on the back of the thumb. The mitten shell is insulated on the back of the hand and thumb as specified in 3.5.4; the palm, back of the hand, gauntlet, and finger/thumb portions are lined with 70D nylon ripstop conforming to 3.5.3. The mitten shell has an elasticized inner wrist and an adjustable strap at the back of the wrist to provide a close fit, with a long gauntlet that extends up the forearm with a manual adjustment feature at the end of the cuff to provide desired closure over various clothing layers. An adjustable lanyard is attached to the hem, and the outside of the shell has a feature at the side seam that enables the mittens to be clipped together as a pair when not in use (see 3.5.13).

3.4.3 Figures. Figures 1 through 4 are furnished for informational purposes only. To the extent an inconsistency exists between the written specification and the figures, the written specification shall govern.

3.5 Materials and components.

3.5.1 Basic shell material. The cloth used to construct the mitten outer shells shall conform to MIL-DTL-32698 (see 6.6.1).

3.5.2 Leather. The leather used to construct the palm, gusset, and a portion of the thumb shall be a 2.25- to 2.75-ounce full grain goatskin conforming to the requirements in table I when tested as specified in 4.4.3. The leather shall be chrome tanned, light finish, fungicide and durable water-repellent treated, and backed by a waterproofing layer. The suede side shall be used to construct a nose wipe on the back of the thumb. The shade of the leather shall be approximate to Coyote 498, conforming to leather shade standard number 3898, and shall be drum dyed, struck through from grain to flesh.

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TABLE I. Leather characteristics and requirements.

Characteristic	Requirement
Thickness, ounces	2.25 to 2.75
Moisture vapor transmission rate (MVTR), grams per square meter in 24 hours (minimum)	500
Water absorption resistance, percent absorption (maximum) 1/	15
Oil and stain resistance, grade (minimum)	4
Abrasion resistance at 1,000 cycles with a CS-10 wheel and 500-gram load	No russet exposure
Stiffness, degrees (maximum) 1/	50
Stitch tear, pounds (minimum) 1/	17
Elongation at 25-pound load, percent (maximum) 1/	25
Shrink temperature, degrees Celsius (minimum)	98±0.5
Veslic rubbing abrasion, rating after 80 cycles (minimum) 2/	4, no damage
Cold crack resistance, visual after exposure at -30 °C	No cracking
Finish adhesion, newtons/10 millimeters, dry and wet (minimum)	4
Flexibility and adhesion of finish at 30,000 cycles after exposure at -20 °F	No cracking or delamination
Hexane soluble material, percent (maximum) 3/	25
Chromic oxide, percent (minimum) 3/	3
pH value (minimum)	3.3
Total ash, percent (maximum) 3/	9
Determination of chromium (VI) content, mg/kg	<3

FOOTNOTES:

1/ Passing values in 80 percent of tested samples required to pass.

2/ Value given using AATCC EP1.

3/ Calculated on moisture-free basis.

3.5.3 Lining material. The lining material used to construct the full inside lining of the mittens shall conform to MIL-DTL-32699 (see 6.6.2).

3.5.4 Insulation. The insulation used on the back of the hand and outer thumb may be sewn to the lining material at the back of the hand as needed to hold it in place. The insulation shall conform to type I, class 2 or 3; type II, class 3; or type III, class 4 of MIL-PRF-32567 (see 6.6.3) with the following exceptions for all classes:

a. The average clo value of all samples prior to the initial laundering shall be a minimum of 2.3 and after laundering shall be a minimum of 2.1, with no individual sample having a clo value measuring more than 10% less than the average.

b. Denier is not required to be as specified.

c. Fibers are not required to be hollow core.

d. Fiber classification is not required to be hollow core, but shall be polyester and one of the following, or a mixture of the following:

(1) Continuous filament

(2) Staple cut fibers

(3) Microfibers

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3.5.5 Elastic. The elastic shall be attached at the inner wrist of the mitten shells using a stretch stitch (such as zigzag) to gather for a closer fit. It shall be 1 inch in width and conform to the requirements of type II, class 1 of MIL-W-5664 (see figures 1 and 3).

3.5.6 Webbing. Webbing shall be used to form a strap at the back of the wrist. Webbing shall be attached at the thumb side seam of the mitten to form a loop for attachment of the buckle (see 3.5.7). The cut length of the loop shall be 3-1/4±1/8 inches long. Another piece of webbing shall be attached at the pinky side outer seam to engage with the buckle to tighten for a closer fit. The cut length of this webbing shall be 11 inches; the end shall be routed through the buckle, folded over 5/8 inch, then folded a second time and top stitched through the three plies 1/2±1/16 inch from the folded end. This shall create a finished loop, impeding the buckle from being removed from the webbing as well as creating a pull tab to tighten for a closer fit. The length of the finished loop shall be 7/8±1/8 inch. The webbing shall be 3/4 inch wide conforming to MIL-PRF-17337. The color shall approximate Coyote 498 or Tan 499 (see figures 2 and 4).

3.5.7 Buckle. The buckle (see 6.6.4) shall engage with the webbing to tighten the mitten at the wrist for a closer fit. It shall be 3/4-inch acetal, and the color shall approximate Coyote 498 (see figures 2 and 4).

3.5.8 Adjustable lanyard. An adjustable lanyard shall be sewn into the pinky side inside lining seam of the shells. The lanyard shall be flat braided nylon tape and conform to A-A-52080, size 1. The lanyard shall be adjustable using a cord lock (see 3.5.11). The ends of the lanyard shall be finished off in a way to restrict the barrel lock from being removed when attached to the mitten shell. The color shall approximate Coyote 498; black may be used as an alternate (see figures 1 and 3).

3.5.9 Thread. The thread for all construction seams shall be nylon, size letter B (Tex size 45), conforming to A-A-59826 type I, class B. The color shall approximate Coyote 498.

3.5.10 Elasticized draw cord. An elasticized draw cord (see 6.6.6 and figures 1 and 3) shall run through the hem of the mitten cuffs and be adjustable using a cord lock (see 3.5.11). The draw cord shall allow for adjustment via a finished off opening, approximately 1/2 inch in length at the pinky side outer seam; the cut length of the cord shall be 22±1/8 inches long. The draw cord shall pull out through the opening and through the cord lock. To restrict the cord lock from being removed from the elasticized draw cord, a cord end shall be used (see 3.5.12). The elastic shall be sufficient for continuous expansion and relaxation, as well as when it is relaxed. The elastic shall allow for comfort on the arm and provide desired closure over various clothing layers. The cord shall be an 1/8-inch elastic cord. The color shall approximate Coyote 498; black may be used as an alternate.

3.5.11 Cord lock. A cord lock (see 6.6.5 and figures 1 and 3) shall be used to allow for adjustment of the lanyard (see 3.5.8) as well as the elasticized draw cord (see 3.5.10). The cord lock shall normally be in the closed position, cinching the cord, when no pressure is on the plunger. For use with the elasticized draw cord, the cord lock shall attach to the pinky side outer seam of the mittens using 3/8-inch nylon tape (see 3.5.14). This attachment shall assist in allowing for one handed adjustment. The color shall approximate Coyote 498 or, as an alternate, may be black.

3.5.12 Cord end. A cord end (see 6.6.7 and figures 1 and 3) shall be used on the elasticized draw cord to restrict the cord lock from being removed. The color shall approximate Coyote 498, black may be used as an alternate.

3.5.13 Clasp. A clasp (see 6.6.8 and figures 2 and 4) shall be attached on opposite sides of the left and right mitten side seams to allow them to be clipped together as a pair. Nylon tape conforming to 3.5.14 shall be used for attachment of the clasp. The color shall approximate Coyote 498, black may be used as an alternate.

3.5.14 Nylon tape. Nylon tape shall be used to secure the cord lock (see 3.5.11 and figures 1 and 3) to the pinky side outer seam of the cuff. The tape shall be nylon, 3/8 inch in width, and conform to type III of MIL-DTL-5038. The color shall approximate Coyote 498; black may be used as an alternate.

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3.5.15 Seam tape (type I only). The full mitten shell shall be seam taped, providing complete water proofing. All seams and stitch lines shall be seam sealed on the inside seams; seam tape shall be centered over the stitch lines. Attachment point tabs shall be created by folding the seam seal tape on itself. The cut length shall be 3 inches to form a 3/4-inch self-faced tab with a 3/4-inch length of single-layer seam seal tape on each side to affix to the specified locations. The mitten shall have a minimum of three attachment point tabs located at the tip of the thumb, the top of the mitten section, and the pinky side of the mitten section.

3.5.16 Labels. Each mitten shall have a combination identification-size label and instruction label. The inscription shall have a minimum font size of 10 points. The inscription legibility, label, and label attachment method shall last the expected life of the mitten. The fastness to laundering requirements of MIL-DTL-32075 shall apply. The labels shall identify the item, boldly specify size, and provide use and care instructions.

3.5.16.1 Identification-size label. The label shall be located inside the mittens on the back of the wrist of both mittens in a pair. The combination identification-size label shall conform to type VI, class 4, MIL-DTL-32075 and shall include the following information, as applicable to the type of shell:

NOMENCLATURE [1/]

SIZE: MEDIUM (EXAMPLE)

CONTRACT NO.:

CONTRACTOR'S NAME:

FIBER CONTENT:

3.5.16.2 Instruction label. The instruction label for the mittens shall conform to type VI, class 3 of MIL-DTL-32075 and shall include the following information:

1. Machine wash on regular cycle in cold water.
2. Do not bleach.
3. Drip dry.
4. Do not iron.
5. Do not dry-clean.

DO NOT REMOVE THIS LABEL

3.5.16.3 Combination identification-size-instruction label. As an alternative to the separate labels in 3.5.16.1 and 3.5.16.2, a combination identification-size-instruction label may be used conforming to type VI, class 14 of MIL-DTL-32075. All information in 3.5.16.1 and 3.5.16.2 shall be included in the combination label.

3.5.16.4 Label/tag. Each mitten shall have an individual bar code placed on a paper tag for personal clothing items conforming to type VIII, class 17 of MIL-DTL-32075. The paper tag shall contain the National Stock Number (NSN) and shall be standard bleached sulfate having a basis weight of 100 pounds. The paper used for the tags shall have a smooth finish to accept thermal transfer and direct printing. The tags shall have a hole and shall be attached to each item by a fastener. The tags shall be clearly legible and readable by a scanner.

FOOTNOTE:

[1/] Nomenclature shall be one of the following, as applicable:

MITTEN SHELL, FULL, EXTREME COLD WEATHER (type I)

MITTEN SHELL, TRIGGER FINGER, EXTREME COLD WEATHER (type II)

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3.6 Patterns. Standard patterns, providing a seam allowance of 1/8 inch for shell and lining seams (with the exception of the portion of the side seam joining the elastic tunnel, at wrist, to the back of the hand, which may require up to 1/4 inch seam allowance to ensure catching all fabric layers) and 1 inch for a hem with drawcord tunnel, will be furnished by the Government (see 6.2). The list of pattern parts and computer nomenclature in table II is provided to ensure that the pattern set provided is complete. The Government patterns shall not be altered in any way, and are to be used only as a guide for cutting the contractor's working patterns. The working patterns shall be identical to the Government patterns, except that additional notching, if needed, to facilitate manufacture may be added. Also, minor modifications may be made where necessary to accommodate the manufacturer's processes and the use of automated equipment. These modifications shall not alter the serviceability or appearance requirements of the end items.

3.6.1 List of pattern parts. The component parts shall be cut from the specified material in accordance with the pattern parts listed in table II.

TABLE II. Pattern parts, types I and II.

Material	Type I	Type II	Nomenclature	Cut Parts 1/
Basic shell material	X		MITTEN-SHELL-BACK_OF_HAND_NYLON	1 left, 1 right
	X		MITTEN-SHELL-CUFF_PALM_NYLON	1 left, 1 right
	X		MITTEN-SHELL-PALM_NYLON	1 left, 1 right
	X		MITTEN-SHELL-INNER_THUMB_NYLON	1 left, 1 right
	X		MITTEN-SHELL-OUTER_THUMB_1_NYLON	1 left, 1 right
	X		MITTEN-SHELL-OUTER_THUMB_2_NYLON	1 left, 1 right
	X		MITTEN-SHELL-INNER_GUSSET_NYLON	1 left, 1 right
	X		MITTEN-SHELL-OUTER_GUSSET_NYLON	1 left, 1 right
	X		MITTEN-SHELL-ELASTIC_COVER_NYLON	1 left, 1 right
		X	TF-MITTEN-SHELL-BACK_OF_HAND_NYLON	1 left, 1 right
		X	TF-MITTEN-SHELL-CUFF_PALM_NYLON	1 left, 1 right
		X	TF-MITTEN-SHELL-PALM_NYLON	1 left, 1 right
		X	TF-MITTEN-SHELL-INNER_THUMB_NYLON	1 left, 1 right
		X	TF-MITTEN-SHELL-OUTER_THUMB_1_NYLON	1 left, 1 right
		X	TF-MITTEN-SHELL-OUTER_THUMB_2_NYLON	1 left, 1 right
		X	TF-MITTEN-SHELL-FINGER_GUSSET_NYLON	1 left, 1 right
		X	TF-MITTEN-SHELL-BACK_OF_TF_NYLON	1 left, 1 right
		X	TF-MITTEN-SHELL-ELASTIC_COVER_NYLON	1 left, 1 right
Leather	X		MITTEN-SHELL-PALM_LEATHER	1 left, 1 right
	X		MITTEN-SHELL-INNER_THUMB_LEATHER	1 left, 1 right
	X		MITTEN-SHELL-INNER_GUSSET_LEATHER	1 left, 1 right
	X		MITTEN-SHELL-OUTER_GUSSET_LEATHER	1 left, 1 right
		X	TF-MITTEN-SHELL-PALM_LEATHER	1 left, 1 right
		X	TF-MITTEN-SHELL-INNER_THUMB_LEATHER	1 left, 1 right
		X	TF-MITTEN-SHELL-FINGER_GUSSET_LEATHER	1 left, 1 right
Leather (suede side)	X		MITTEN-SHELL-OUTER_THUMB_1_SUEDE	1 left, 1 right
		X	TF-MITTEN-SHELL-OUTER_THUMB1_SUEDE	1 left, 1 right

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TABLE II. Pattern parts, types I and II – Continued.

Material	Type I	Type II	Nomenclature	Cut Parts 1/
Lining material	X		MITTEN-SHELL-BACK_OF_HAND_RIPSTOP	1 left, 1 right
	X		MITTEN-SHELL-PALM_RIPSTOP	1 left, 1 right
	X		MITTEN-SHELL-OUTER_THUMB_RIPSTOP	1 left, 1 right
	X		MITTEN-SHELL-INNER_THUMB_RIPSTOP	1 left, 1 right
		X	TF-MITTEN-SHELL-BACK_OF_HAND_RIPSTOP	1 left, 1 right
		X	TF-MITTEN-SHELL-PALM_RIPSTOP	1 left, 1 right
		X	TF-MITTEN-SHELL-OUTER_THUMB1_RIPSTOP	1 left, 1 right
		X	TF-MITTEN-SHELL-OUTER_THUMB2_RIPSTOP	1 left, 1 right
		X	TF-MITTEN-SHELL-INNER_THUMB_RIPSTOP	1 left, 1 right
Insulation	X		MITTEN-SHELL-BACK_OF_HAND_INSULATION	1 left, 1 right
	X		MITTEN-SHELL-OUTER_THUMB_INSULATION	1 left, 1 right
		X	TF-MITTEN-SHELL-BACK_OF_HAND_INSULATION	1 left 1 right
		X	TF-MITTEN-SHELL-OUTER_THUMB1_INSULATION	1 left, 1 right
		X	TF-MITTEN-SHELL-BACK_OF_TF_INSULATION	1 left, 1 right

FOOTNOTE:

1/ Pieces shall be cut face-to-face to create left and right cut parts.

3.7 Construction. The assembled mitten shells (referred to as “end item” in this specification) shall be as specified in table III to ensure environmental protection, comfort, and durability in garrison use, combat, and field operations. End item construction and appearance shall conform to the requirements of this document, end item characteristics specified in table III, and the finished dimensions as specified in 3.9 to maintain item configuration and compliance to end item and component performance tests (see 4.4 and 4.5).

TABLE III. End item requirements, types I and II.

Characteristic	Requirement
Clo (dry thermal insulation) (minimum) value 1/	2.2
Liner retention, pounds (minimum)	6
Overall liquid integrity	Pass

FOOTNOTE:

1/ Testing will be performed in combination with the insulated removable liner conforming to MIL-DTL-32708. Requirement listed is the system requirement.

3.7.1. Stitches, seams, and stitching. All seams shall exhibit a uniform appearance and conform to ASTM D6193. Stitch types and stitches per inch shall be as listed in table IV. All material edges shall not ravel. Edges may be turned-in, turned-under, or serged to prevent raveling.

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TABLE IV. Stitch types and stitches per inch.

Operation	Stitch Type	Stitches Per Inch
Topstitch, single needle stitch, or straight stitch	301	8 to 12
Three-step zigzag	315	8 to 12
Baste	301 or 101	8 to 12

3.7.2 Tacking and backstitching. The inside lining shall be tacked to the shell at several locations to ensure the lining material does not separate from the shell. This shall be accomplished by using 70D nylon ripstop to create a minimum of four attachment point tabs located at the tip of the thumb and the tip of the trigger finger, in addition to the top and pinky side of the finger portion of the mitten. These tabs shall then be tacked to the seam allowance of the liner. Ends of seams and rows of stitching with 301-type stitch, when not caught in other seams or stitching, shall be securely backstitched not less than 1/4 inch. Thread breaks (all stitch types) shall be secured by stitching back from the break not less than 1/2 inch.

3.8 Color matching.

3.8.1 Visual shade matching. The color and appearance of the materials and components shall approximate the standard sample when tested in accordance with 4.5.5.

3.8.2 Color. The exterior mitten materials shall be solid and shall closely approximate Coyote 498 shade, which complies with the AMS-STD-595 chip 20150. The exterior materials and components shall meet the reflectance requirements specified in 3.8.2.1.

3.8.2.1 Spectral reflectance. The finished outer shell cloth and all components used shall meet the spectral reflectance values (in percent) for the visible/near infrared wavelength range, 600 to 860 nanometers for the colors specified in table V as applicable when tested as specified in 4.5.4.

TABLE V. Spectral reflectance.

Wavelengths (nanometers)	Coyote 498	
	Minimum	Maximum
600	8	20
620	8	20
640	8	22
660	8	24
680	12	24
700	12	34
720	16	42
740	22	46
760	30	50
780	34	54
800	36	56
820	38	58
840	38	58
860	40	60

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3.9 Finished measurements. Finished dimensions shall be in accordance with tables VI and VII and measured in accordance with table VIII. Measurements should be taken with the wrist strap loosened enough that it does not affect the measurements.

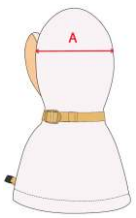
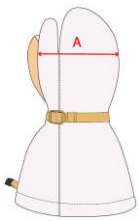
TABLE VI. Mitten shell finished dimensions, type I (in inches).

Size and Tolerance	A: Back of Hand Width	B: Full Length	C: Hand Length	D: Wrist Width, Relaxed	E: Thumb Length	F: Thumb Width
Small	5-1/2	16-3/8	8	5	2-7/8	2-1/2
Medium	5-3/4	16-7/8	8-1/4	5-1/4	3-1/8	2-1/2
Large	6	17-3/8	8-1/2	5-1/2	3-3/8	2-1/2
Extra Large	6-1/4	17-7/8	8-3/4	5-3/4	3-5/8	2-1/2
Tolerance	1/4	1/4	1/4	1/4	1/8	1/8

TABLE VII. Mitten shell finished dimensions, type II (in inches).

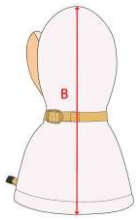
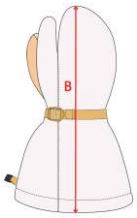
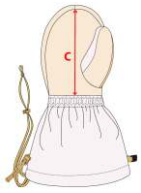
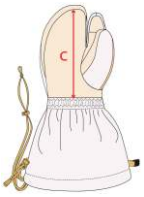
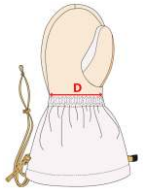
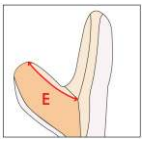
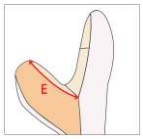
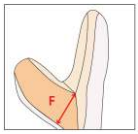
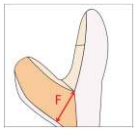
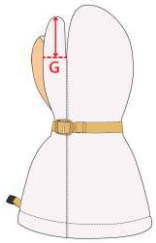
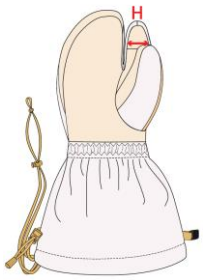
Size and Tolerance	A: Back of Hand Width	B: Full Length	C: Hand Length	D: Wrist Width, Relaxed	E: Thumb Length	F: Thumb Width	G: First Finger Length	H: First Finger Width
Small	6-3/8	16-5/8	8-1/2	4-3/4	2-7/8	2-1/2	3-3/4	1-3/4
Medium	6-5/8	17-1/8	8-3/4	5	3-1/8	2-1/2	3-3/4	1-7/8
Large	6-7/8	17-5/8	9	5-1/4	3-3/8	2-1/2	3-3/4	2
Extra Large	7-1/8	18-1/8	9-1/4	5-1/2	3-5/8	2-1/2	3-3/4	2-1/8
Tolerance	1/4	1/4	1/4	1/4	1/8	1/8	1/8	1/8

TABLE VIII. Mitten shell methods of measurement.

Measurement Name	Measurement Description	Figure	
		Type I	Type II
A: Back of hand width	Type I: Measurement taken on the back of the hand with adjustable strap loosened. Measurement parallel to the mitten shell opening across the widest part of the mitten (see note), beginning and ending at the seams joining the shell material to the leather fourchette. Type II: Measurement taken on the back of the hand with adjustable strap loosened. Measurement parallel to the mitten shell opening across the widest part of the mitten (see note), edge-to-edge, with the mitten laid flat. Note: the widest part of the mitten (both types) is even with the thumb crotch seam.		

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TABLE VIII. Mitten shell methods of measurement – Continued.

Measurement Name	Measurement Description	Figure	
		Type I	Type I
B: Full length	(Both types) Measurement taken on the back of the hand at the center of the mitten from the top seam (where joining leather fourchette seam meets shell material) to the bottom of the cuff.		
C: Hand length	Measurement taken on the palm side from the top seam (where the leather palm seam joins the leather fourchette) to the bottom of the leather palm (seam joining the leather palm to shell fabric at the wrist).		
D: Wrist width, relaxed	Measurement taken on the palm side, following the zigzag seam closest to the leather palm.		
E: Thumb length	Measurement taken from the thumb tip following suede to the leather seam ending at the bottom of the seam at thumb crotch.		
F: Thumb width	Measurement taken across the back of the thumb suede, parallel to the thumb tip, from the intersection of the thumb crotch seam to the edge where suede meets the shell.		
G: First finger length	Measurement taken on the back of the hand at the center of the finger beginning at the fingertip seam down to the finger crotch seam.	N/A	
H: First finger width	Measurement taken across the front of the first finger (from seam to seam where the shell material meets leather), parallel to the fingertip, at dart.	N/A	

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3.10 Workmanship. The finished mitten shells shall be free from loose thread, foreign matter, and irregular defects that can adversely affect usage or durability. The finished items shall be uniform in quality, free from defects that adversely affect form, fit, or function, and the defects specified in 4.4.1. The occurrence of defects shall not exceed the established acceptance quality limits (AQL) (see 6.2) and shall not adversely affect the serviceability, appearance, and uniformity of the product.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. A first article inspection shall be performed on the mitten shell when a first article sample is required (see 6.2). This inspection shall consist of the examinations and tests specified in 4.4. The first article inspection shall be considered a failure, and the first article sample shall be rejected, if any associated requirements are not met.

4.2.1 First article samples. The sample unit shall be one pair of mitten shells and the lot size shall be expressed in pairs. The first article sample size shall be as specified in the procurement document (see 6.2).

4.3 Conformance inspection. Conformance inspection shall include the examination specified in 4.3.1.

4.3.1 Conformance inspection samples. Sampling for conformance inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with AQLs as specified in the contract or order (see 6.2). The sample unit shall be one pair of mitten shells and the lot size shall be expressed in pairs.

4.3.2 In-process examination. Visual and dimensional examinations shall be made at any point or during any phase of the manufacturing process to determine whether the construction details that cannot be easily examined in the finished product are in accordance with requirements specified in section 3. Materials and components that contain a defect in accordance with table IX shall be corrected or removed from production.

4.4 Component and end item inspection. In accordance with 4.1 and unless otherwise excluded, amended, or modified in this specification or applicable procurement document (see 6.2), components and end items shall be inspected in accordance with all the requirements of the referenced documents. All requirements are applicable to the sample unit. All test reports should contain the individual values utilized in expressing final results. The lot shall be rejected if one or more sample units fail to meet any test requirement specified.

4.4.1 End item visual examination. The mitten shell shall be visually examined for defects in shade, design, material, construction, and workmanship, with defects classified in accordance with table IX. Defects shall not exceed the AQL specified in the contract or order (see 6.2).

TABLE IX. Defects.

Examination	Defect	Major	Minor
Mitten pairing	Not matched and paired, right and left not same size, or wide variation in appearance	101	
	Not tacked or joined together as specified		201
Design	Not as specified (incorrect material, pattern, etc.)	102	
	Poorly shaped		202
	Buckles or hardware positioned incorrectly	103	
Shade	Shade variation within a part or between parts	104	
	Unacceptable shade change as a result of water repellent treatment application	105	
	Any color not as specified	106	

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TABLE IX. Defects – Continued.

Examination	Defect	Major	Minor
Cleanness	Spot, stain		203
	Excessive thread ends not trimmed or removed		204
	Shade stamp marking on outside		205
	Odor, affecting appearance or serviceability		206
Material and workmanship	Any component part omitted, distorted, full, tight, or twisted	107	
	Not functioning properly or defective	108	
	Any part caught in any unrelated stitching	109	
	Cloth material:		
	Any hole, cut, tear, smash, burn, drill hole, run, thin place, dye streak, misweave, knot, or slub affecting appearance or serviceability	110	
	Shell material not properly backed with waterproofing layer	111	
	Leather:		
	Not specified color	112	
	Not full grain	113	
	Flesh side not smooth or contains area of coarse, loose fibers	114	
	Not clean — stained or foreign matter		207
	Hard, boney, loose, spongy leather, or hard scar	115	
	Cut or hole	116	
	Seam (defect affecting appearance or serviceability):		
	Not specified type	117	
	Puckered, distorted, pleated, wavy, twisted, irregular, or open	118	
	Loose or tight stitch tension, broken or missing thread or stitch	119	
	Needle cutting, needle chew, or visible mend	120	
	Edge or raised stitching sewn too close to edge resulting in damage to cloth	121	
	Seam allowance not as specified	122	
	Visible raw edge	123	
	Any visible perforations from removed stitches		208
	Bartacks missing, insecure, misplaced, not specified size; stitches loose or broken; or bartack not serving intended purpose	124	
	Front and back of mitten uneven by more than 1/2 inch at hem	125	
	Unwanted permanent fold, pleat, or crease in fabric or mitten, affecting appearance or serviceability	126	
Labels	Omitted, incorrect, illegible, or not attached where specified		209
	Barcodes omitted or not readable by scanner		210
	Human-readable interpretation (HRI) omitted or illegible		211
	Barcode not visible on folded, packaged item; bar code attachment causes damage to the item		212
Packaging	Not packaged in accordance with the contract or purchase order (see 6.2)		213

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4.4.2 Dimensional examination. The end item shall be examined for conformance to the dimensions specified in table VI and table VII.

4.4.3 Leather testing. The leather (see 3.5.2) shall be tested for the characteristics listed in table X. The methods of test shall be as specified in table X for the requirements in table I. The testing requirements and test sampling plan shall be as specified in the contract or purchase order (see 6.2).

TABLE X. Leather tests.

Characteristic	Test Method
Thickness	ASTM D1814
MVTR	ASTM D5052
Water absorption resistance	see 4.5.1
Oil and stain resistance	AATCC TM118
Abrasion resistance	ASTM D7255
Stiffness	ASTM D2821
Stitch tear	ASTM D4705
Elongation	ASTM D2209
Shrink temperature	ASTM D6076
Veslic rubbing abrasion	ISO 11640
Cold crack resistance	ASTM D1912
Finish adhesion	ISO 11644
Flexibility and adhesion of finish	ASTM D6182
Hexane soluble material	ASTM D3495
Chromic oxide	ASTM D2807
pH value	ASTM D2810
Total ash	ASTM D2617
Determination of Chromium (VI) content	ISO 17075-2

4.4.4 End item testing. The end item shall be tested for the characteristics listed in table XI. The methods of test shall be as specified in table XI for the requirements in table III.

TABLE XI. End item testing.

Characteristic	Verification
Clo (dry thermal insulation) value	see 4.5.2
Liner retention	see 4.5.3
Overall liquid integrity	NFPA 1971 (see 6.7)

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4.5 Examinations and tests.

4.5.1 Water absorption resistance test. One 3-inch diameter specimen shall be tested from each 12-by-12-inch test area. Prior to testing, the specimen shall be allowed to reach moisture equilibrium in the standard atmospheric conditions environment, as specified in ASTM D1610. A stainless steel 500-milliliter, 3-inch diameter, 5-inch long container with a rubber gasket and cover shall be used to test the specimen. The specimen shall first be weighed (to the nearest 0.01 gram) and then shall be placed inside the cover with the flesh side of the specimen next to the inside of the cover. The rubber gasket shall then be placed inside the cover and against the grain surface of the leather specimen. One-hundred milliliters of distilled or deionized water at 23 °C shall then be added to the container. The cover with the specimen and gasket shall then be secured to the container. The container shall then be inverted so the water is against the grain surface of the specimen. The inverted container shall be placed on a flat surface and allowed to remain in the inverted position for 30 minutes. At the end of 30 minutes, the container shall be placed upright and the cover shall be removed and placed on a flat surface. The inside of the cover shall face the flat surface. The cover shall be tilted so the outer edge of the bottom is raised 1 to 1-1/2 inches from the flat surface and allowed to drain for 5 minutes. After 5 minutes, the specimen shall be removed from the cover, held by its edge, and shook briskly once to remove any free moisture from the surface of the specimen. The specimen shall be immediately reweighed to the nearest 0.01 gram; the percent water absorption shall be calculated as follows:

$$\text{Percent water absorption} = \frac{(W2 - W1) \times 100}{W1}$$

Where:

W1 = original weight of conditioned specimen

W2 = weight of specimen after being subject to water

4.5.2 Thermal insulation test. Three separate samples of the mitten shells shall be evaluated in accordance with ASTM F3426 and the average value shall be reported. Total thermal resistance (R) values can be converted to the more familiar clo unit by multiplying the R value by 6.45 to allow ranking order of standard for procurement purposes. Test size shall be right hand, size large. A life-sized biophysical model of the human hand that measures localized and total thermal resistance shall be used. Insulation values for the total hand model and its seven individual regions are calculated by an internal program during 30 minutes of steady state data collection. The ambient temperature shall be maintained at 5±0.5 °C and the surface of the hand model is controlled at 35 °C. The relative humidity shall be 50 percent and wind speed shall be maintained at 0.4 meters per second. Total thermal resistance (R) to heat exchange is calculated in the test conditions stated above and using:

$$R = \frac{(SA \times \text{delta T})}{P}$$

Where:

R = the thermal resistance, m^2 °C/watt

SA = total surface area of seven regional segments, m^2

delta T = temperature gradient between the hand model surface and ambient air temperature, °C

P = Total power input, watts

If required (see 6.2), mittens submitted will be tested by the Government. Testing at alternate facilities shall require side-by-side comparison due to chamber wind speed differences that cannot be controlled and that will affect the total measured values.

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4.5.3 Liner retention test and testing apparatus. Liner retention shall be evaluated with the use of locking forceps and a force-measuring gauge. A minimum of three finished items shall be tested. Each digit of the mitten shell shall be tested as follows. The locking forceps shall be attached to the inner liner of the digit to be tested ensuring that an unattached liner or the outer shell is not grabbed. The hook of the force gauge shall be looped around the locking bridge of the forceps. The digit of the mitten shell shall be gripped ensuring the inner liner is not impeded. The force gauge shall be pulled until the 6 pound (threshold requirement) and 8 pound (objective requirement) registers on the dial and then released. Each digit shall be inspected for indication of detachment of inner barrier liner. Results shall be recorded and reported as pass or fail. Failure of any attachment point of any mitten shell shall constitute failure. Items shall be permitted to be cut open to verify detachment.

4.5.4 Spectral reflectance test. Spectral reflectance data shall be obtained from 600 to 860 nanometers at 20-nanometer intervals on a spectrophotometer relative to the polytetrafluoroethylene (PTFE) family of compounds, the preferred white standard. Other white reference materials may be used provided they are calibrated to absolute white or vitrolite tiles. The spectral band width shall be less than 20 nanometers at 860 nanometers. Reflectance measurements shall be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode of operation is used, the spectrophotometer shall operate with the specimen diffusely illuminated with the full emission of a continuous source that simulates either International Commission on Illumination (CIE) source A or CIE source D65. Measurements shall be taken on a minimum of two different areas and the data averaged. The measured areas should be at least 6 inches away from the selvage. The specimen shall be measured as a single layer backed with three layers of the same fabric and shade. The specimen shall be viewed at an angle no greater than 10 degrees from normal, with the specular component included. Measurements shall be taken on a minimum of two different areas. Specimens shall be oriented in different directions during testing. When possible, the specimens tested shall not contain the same warp or filling yarns or wales or courses when presented to the sample port. Camouflage materials should be measured with the appropriate aperture size to ensure that only one color is measured at a time. Always try to use the largest aperture possible. Photometric accuracy of the spectrophotometer shall be within 1 percent and wavelength accuracy within 2 nanometers. Any color having spectral reflectance values falling outside the limits at four or more of the wavelengths specified shall be considered a test failure.

4.5.5 Visual shade matching. Samples shall be viewed using AATCC EP9, option C (see 6.5), with sources simulating artificial daylight D65 illuminant (see 6.5.1) with a color temperature of 6500±200 Kelvin (K) illumination of 100±20 foot candles, and shall be a good approximation to the standard sample under incandescent A illuminant with a color temperature of 2856±200K.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The mitten shells are intended for wear by military personnel of the Department of Defense during extreme cold weather conditions in combination with a mitten liner conforming to MIL-DTL-32708.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and size required (see 1.2).

d. Applicable Government patterns, including revisions.

e. When first article sample is required (see 3.1.1, 4.2, and 6.3).

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f. When conformance inspection is required (see 3.1.2).

g. AQLs (see 3.10, 4.3.1, and 4.4.1).

h. Number of first article inspection samples (see 4.2.1).

i. When end item testing may be other than as specified in referenced documents (see 4.4).

j. Leather testing requirements and sampling plan (see 4.4.3).

k. If the mitten shells require testing by the Government (see 4.5.2).

l. Packaging requirements (see 5.1).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory, and the number of items to be tested as specified in 4.2. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product that has been previously acquired or tested by the Government and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Standard sample. For standard samples, address the procuring activity issuing the invitation for bids or request for proposal.

6.5 Visual shade matching. In 2019, option A of AATCC EP9 was changed to option C. NOTE: In case of confusion, the viewing geometry should be such that the specimen plane and illumination source are parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35±5-degree angle relative to the horizontal. The observer will view the specimens at a 90-degree angle, relative to the plane of the specimens.

6.5.1 D75 illuminant. D75 illuminant light source may be used in lieu of D65 light source.

6.6 Sources of supply.

6.6.1 Basic material. A known source that meets the requirements of the material specified in 3.5.1 is the following:

Brookwood Companies, Inc.

485 Madison Avenue, Suite 500

New York, NY 10022

Phone: 212-551-0100

Fax: 646-472-0294

https://www.brookwoodcos.com/

6.6.2 Lining material, gauntlet. A known source that meets the requirements of the material specified in 3.5.3 is the following:

Brookwood Companies, Inc.

485 Madison Avenue, Suite 500

New York, NY 10022

Phone: 212-551-0100

Fax: 646-472-0294

https://www.brookwoodcos.com/

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6.6.3 Insulation. Known sources that meet the requirements of the material specified in 3.5.4 are as follows:

3M	PrimaLoft	Climashield
3M Center	19 British American Blvd	641 Northpark Drive
St. Paul, MN 55144	Latham, NY 12110	Clinton, TN 37716
Phone: 888-364-3577	Phone: 518-213-9073	Phone: 865-463-6184
www.3m.com	www.primaloft.com	Fax: 865-463-6188
		www.climashield.com

6.6.4 Buckle. High release lock, style #94579 HRL has been shown to meet the requirements specified in 3.5.7. A known source of this material is the following:

American Cord & Webbing Co., Inc
88 Century Dr.
Woonsocket, RI 02895
Phone: 401-762-5500
https://acw1.com/

6.6.5 Cord lock. Toaster Ellipse Cordloc, 350-2000 has been shown to meet the requirements specified in 3.5.11. A known source of this component is the following:

ITW Nexus
21601 South Harlem Avenue
Frankfort, IL 60423
Phone: 708-720-3025
Fax: 708-720-2612
http://www.itwnexus.com/

6.6.6 Elasticized draw cord. One-eighth-inch Elastic Cord, Part #CORD-SHK-125-DLK has been shown to meet the requirements specified in 3.5.10. A known source of this component is the following:

Hi-Tex Corporation
16110 Woodinville-Redmond Road NE
Woodinville, WA 98072
Phone: 800-286-0659
Fax: 425-488-3567
info@hitexcorp.com

6.6.7 Cord end. Zipper pull, style #ZP Wide 99540 has been shown to meet the requirements specified in 3.5.12. A known source of this component is the following:

American Cord & Webbing Co., Inc
88 Century Dr.
Woonsocket, RI 02895
Phone: 401-762-5500
https://acw1.com/

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6.6.8 Clasp. Center Release Buckle, Part #1002-0038 has been shown to meet the requirements specified in 3.5.13. A known source of this component is the following:

ITW Nexus

21601 South Harlem Avenue

Frankfort, IL 60423

Phone: 708-720-3025

Fax: 708-720-2612

http://www.itwnexus.com/

6.7 NFPA 1971 Standard. It is recommended that the 2000 edition, Chapter 6-33, "Overall Liquid Integrity Test One" for the cited test (see table XI), as that was the procedure used during the development of this item.

6.8 Subject term (key word) listing.

Handwear

Liner

Taslan

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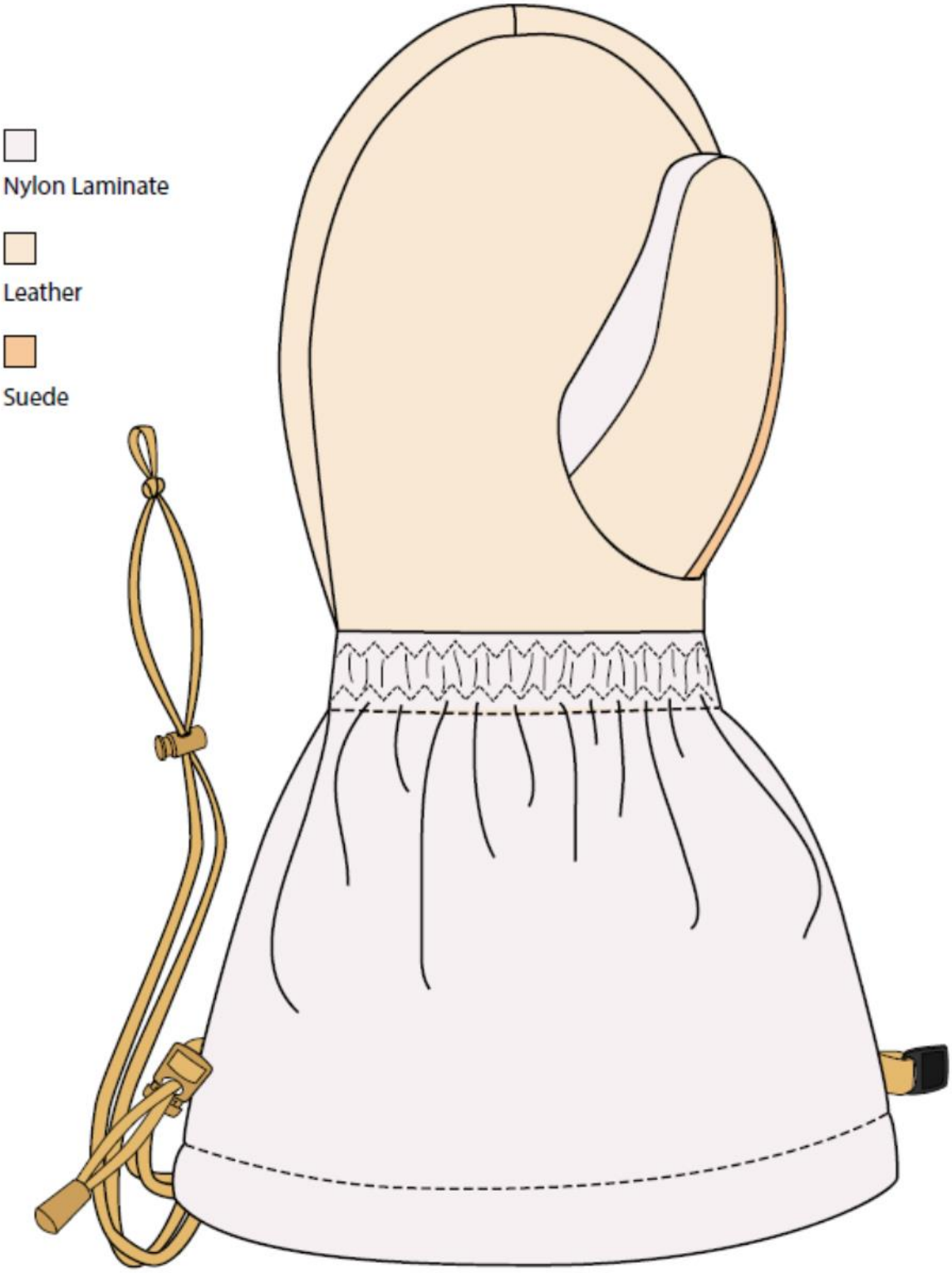


FIGURE 1. Palm view, type I.

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Check the source to verify that this is the current version before use.

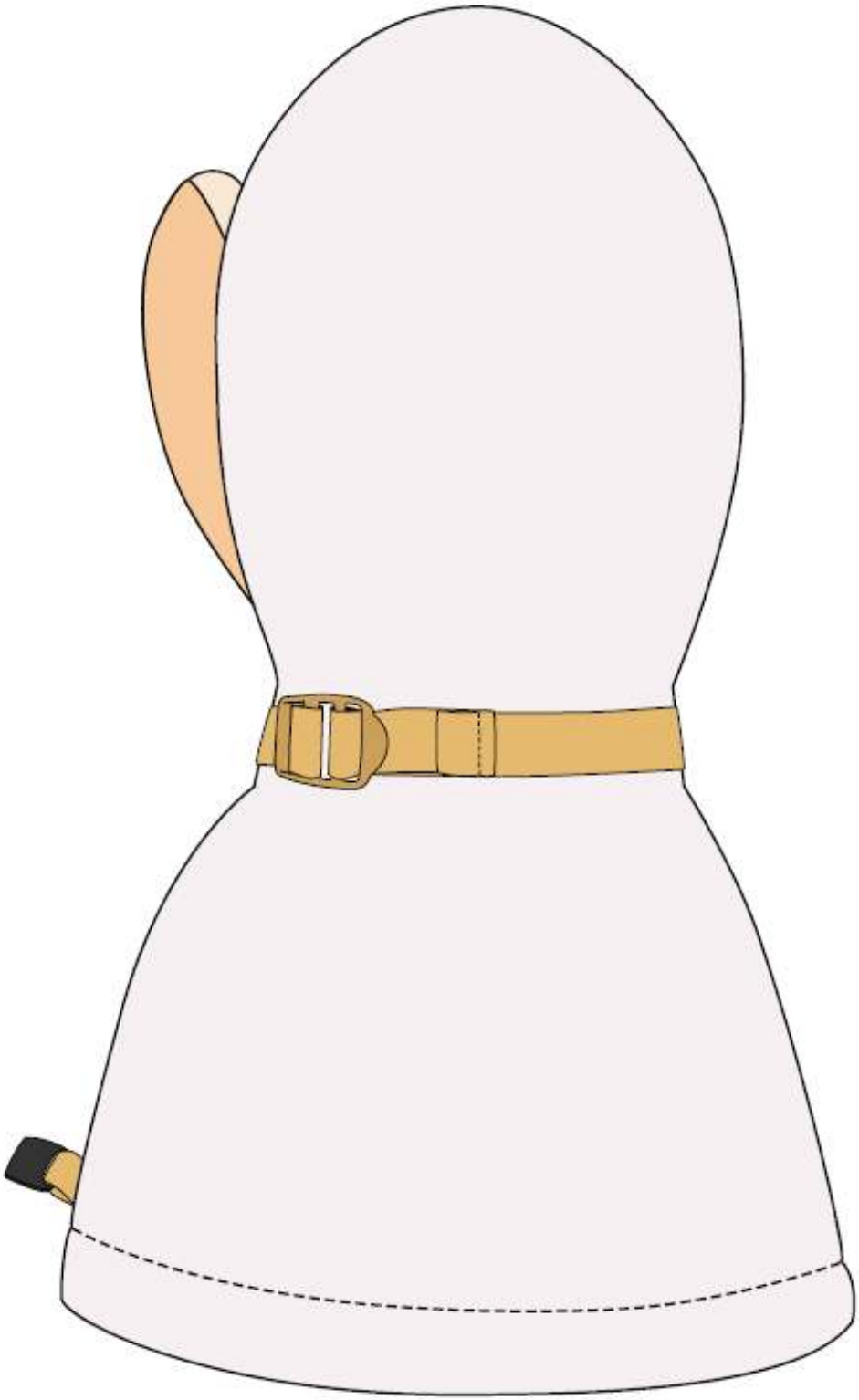


FIGURE 2. Back of hand view, type I.

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Check the source to verify that this is the current version before use.

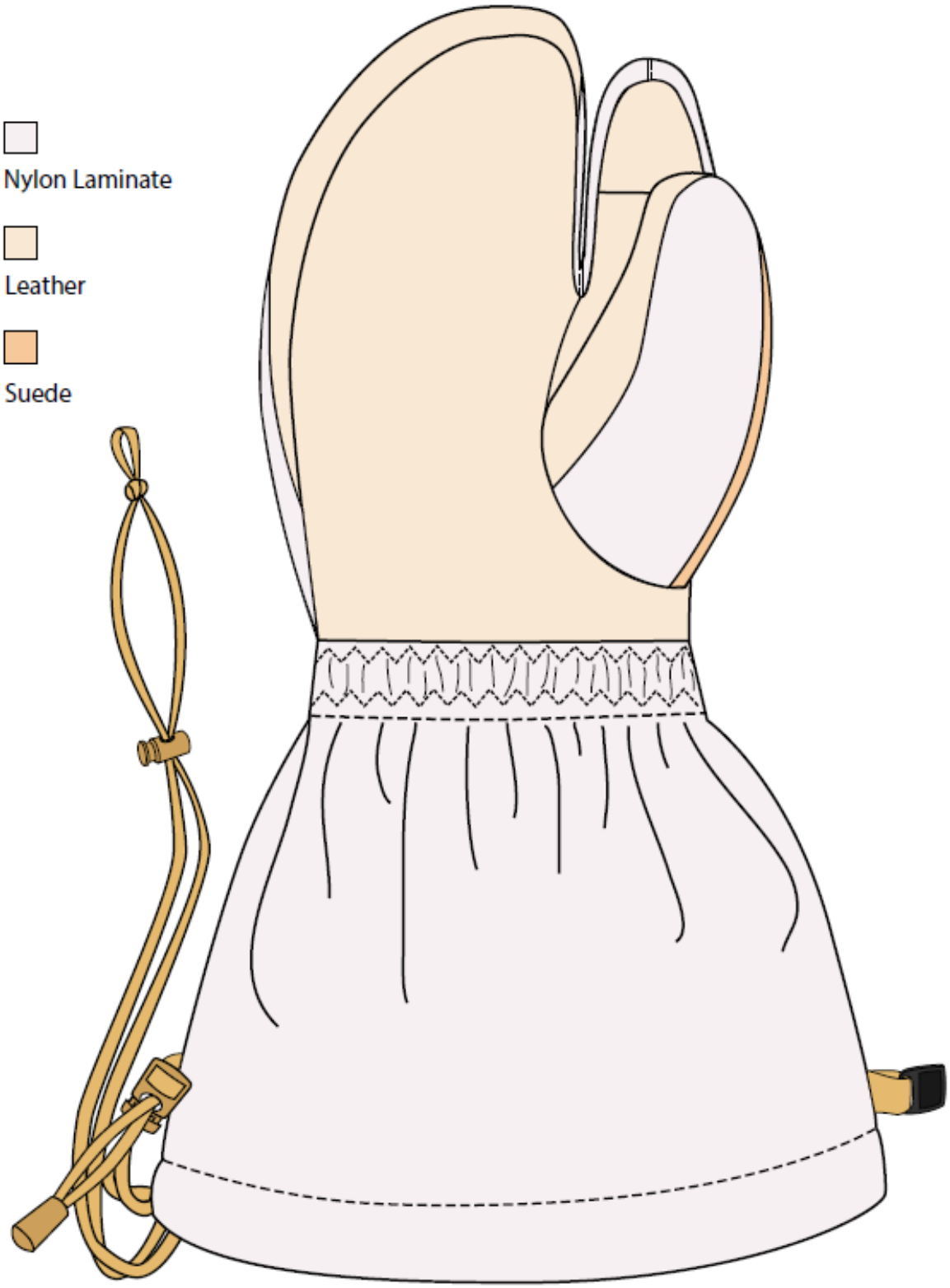


FIGURE 3. Palm view, type II.

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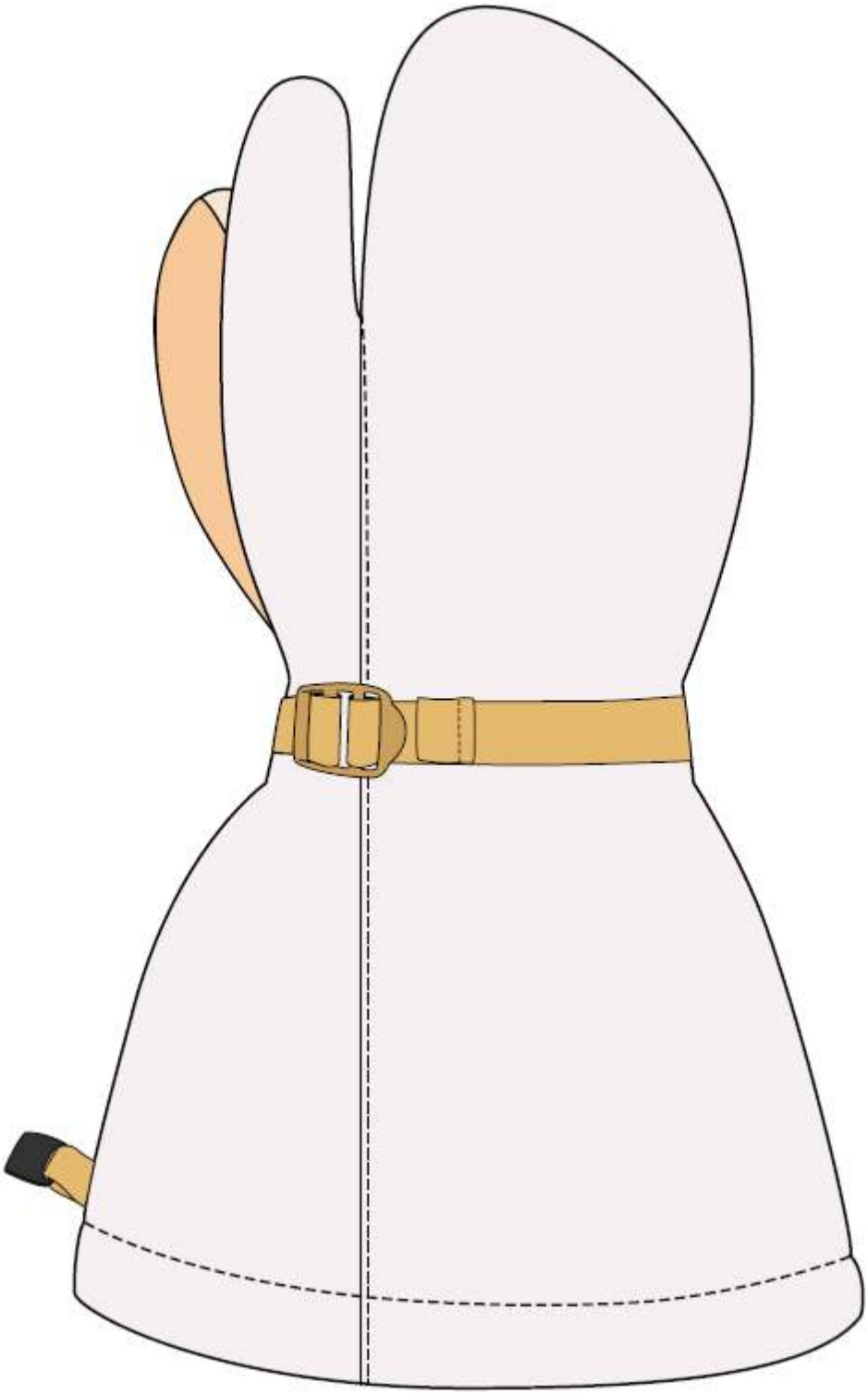


FIGURE 4. Back of hand view, type II.

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CONCLUDING MATERIAL

Custodians:
Army – GL
Navy – MC

Preparing activity:
Navy – MC
(Project 8440-2022-003)

Review activities:
Army – AV, MI
Navy – AS, CG1, NU
Air Force – 11
DLA – CT

Civil agency:
GSA – FAS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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